Q1: Convert Clock Cycle to Clock Rate: 333 x 10-12 seconds

Q2: Given that variables base is in register X6 and variable total will be saved in register X5. Which of the ARMv8 instruc"on sequence represents the following statement?
total = base * 15 + 27

Q3: Which of the following code segments math this 'C' code
seconds = hours * 3,600 + min * 60
X10 Contains the number hours
X11 Contains the number of minutes
X12 will contain the number of seconds

Q4: Given that variables a, b, c, d are in registers X10, X11, X12, X13 respec"vely
Variable g will be saved in register X20.
Which of the ARMv8 instruc"on sequence represents the following statement?
g = (a + b) * (c - d)

Q5: Given that variables base is in register X6
and variable total will be saved in register X5
Which of the ARMv8 instruc"on sequence represents the following statement?
total = base * 15 + 27

Q6: Which of the following code segments math this 'C' code
seconds = hours * 3,600 + min * 60
X10 Contains the number hours
X11 Contains the number of minutes
X12 will contain the number of seconds

Q7: Consider the following LEGv8 loop:

```
        SUB X0, X0, X0
        ADDI X1, XZR, #10
LOOP: SUBIS X1, X1, #0
        B.LE DONE
        SUBI X1, X1, #1
        ADDI X0, X0, #2
```

```
        B LOOP
DONE:
```

What is the final value in register X0?

Q8: Consider the following LEGv8 loop:

```
        SUB X0, X0, X0
        ADDI X1, XZR, #5
LOOP: SUBIS X1, X1, #0
        B.LE DONE
        SUBI X1, X1, #1
        ADD X0, X0, X1
        B LOOP
DONE:
```

What is the final value in register X0?

Q9: Consider the following LEGv8 loop:

```
        SUB X0, X0, X0
        ADDI X1, XZR, #10
LOOP: SUBIS XZR, X0, X1
        B.GT DONE
        SUBI X1, X1, #1
        ADDI X0, X0, #2
        B LOOP
DONE:
```

What is the final value in register X0?

Q10:

1.) Convert -100 to 16-bit Binary. (answer will contain 16 bits)

2.) Convert 0000 1101 0000 0101 to Decimal.

3.) Convert (assume we have an 8-bit SIGNED integer) Binary 1000 0011
To SIGNED Decimal:

4.) Convert (assume we have an 8-bit UNSIGNED integer) Binary 1000 0011
To UNSIGNED Decimal:

Q11:

What is this number in dec 11111001111_2

Covert this base 10 number 1341_{10} to HEX

Convert -451_{10} to binary

Convert -451_{10} to 16-bit HEX